

Yang Zhongyu

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EDUCATION

• Lanzhou University (Project 985)

Sept. 2021 - June.2025

B.S. in Mathematics(the Basic Theory Class)(Main major) and Management (Minor)

Lanzhou, China

- Relevant courses: Mathematical Analysis, Advanced Algebra, Python, C++, Probability Theory, Ordinary Differential Equations, Numerical Analysis, Microeconomics, Differential Geometry, Functional Analysis, etc.
- Thesis: Adaptive Multi-task Medical Image Restoration, Supervised by Prof.Yu-Mei Huang and rated as A⁺.

PATENTS AND PUBLICATIONS

J=Journal, P=Patent, S=Software, T = Tech Report, R=Under Review

* means Corresponding Author, † means equal contribution

- [J.1] Zhongyu Yang, Ziyue Xue. **Analysis and Forecast of GDP of Gansu Province based on ARIMA Model**. *Chinese Market* (IF=0.6)
- [J.2] Mengying Su, Zhongyu Yang*, Shujaat Abbas, et al. **Toward Enhancing Environment Quality in OECD Countries: Role of Municipal Waste, Renewable Energy, Environment Innovation and Environmental Policy**. *Renewable energy* (SCI Q1 Top, IF=9.0)
- [J.3] Zhichao Yu, Wenlan Xie, Junjie Guo, Zhongyu Yang*. **Green Effect of Energy Transition Policy: A quasi-natural Experiment Based on New Energy Demonstration Cities**. *Finance Research Letters* (SSCI Q1 Top, IF=10.4)
- [J.4] Zhongyu Yang†, Dannong Xu†, Wei Pang, Yingfang Yuan. **Script: Graph-Structured and Query-Conditioned Semantic Token Pruning for Multimodal Large Language Models**. *TMLR* 2025.
- [C.1] Zhongyu Yang†, Jun Chen†, Dannong Xu, et al. **WikiAutoGen: Towards Multi-Modal Wikipedia-Style Article Generation**. *ICCV 2025*, Hugging Face Daily Selection.
- [C.2] Zhongyu Yang, Junhao Song, Zuhao Yang, Wei Pang, Yingfang Yuan, et al. **ReChar: Revitalizing Characters with Structure-Preserved and User-Specified Aesthetic Enhancements**. *Siggraph Asia* 2025.
- [C.3] Zhongyu Yang, Junhao Song, Siyang Song, Wei Pang, Yingfang Yuan. **MERMAID: Multi-perspective Self-reflective Agents with Generative Augmentation for Emotion Recognition**. *EMNLP* 2025.
- [P.1] Zhongyu Yang. **A mathematics teaching system based on virtual reality**. (CN116312091A)
- [S.1] Zhongyu Yang. **Green and Low-carbon Integrated Monitoring Software**.(2023SR1355487)
- [S.2] Zhongyu Yang. **Fully automatic spatial sound field environment perception system**. (2024SR0538446)
- [T.1] Zhongyu Yang, Hao Wu. **Tropical Linear Representation of Involute Chinese Monoids**. Technology Report.
- [R.1] Zhongyu Yang†, Yingfang Yuan†, Xuanming Jiang, Baoyi An, Wei Pang. **InEx: Hallucination Mitigation via Introspection and Cross-Modal Multi-Agent Collaboration**. Under review in *AAAI* 2025.
- [R.2] Dannong Xu†, Zhongyu Yang†, Jun Chen, et al. **MultiHaystack: Benchmarking Multimodal Reasoning over 40K Images, Videos, and Documents**. Under review in *ICLR* 2025.

EXPERIENCE

• SenseTime Research

Shenzhen, China

Research Intern, General Perceptual Computing Group

Feb. 2025 – Present

- Research on streaming MLLMs, focusing on hallucination mitigation and consistent vision-text alignment.

• KAUST

Saudi Arabia

Remote Research Intern, Advisor: Dr.Jun Chen and Prof.Mohamed Elhoseiny

Dec. 2024 – Present

- Exploring multi-agent coordination for web-scale retrieval to augment the reasoning capability of MLLMs.

• Heriot-Watt University

Edinburgh, UK

Remote Research Intern, Advisor: Prof.Wei Pang and Dr.Yingfang Yuan

March. 2024 – Present

- Research on diffusion-based image generation and multi-agent systems for task-specialized visual synthesis.

• The Chinese University of Hong Kong, Shenzhen

Shenzhen, China

Research Assistant, Advisor: Prof.Zhang Ruimao

Apr. 2024 – Nov. 2024

- Developed MLLMs' alignment between visual and textual cues to enable human-centric multimodal generation.

• iFLYTEK Co., Ltd.

June 2023 - Aug. 2023

Data Analysis Assistant Intern

Lanzhou, China

- Developed multimodal recommendation models from multi-source behavior-aware cues.

HONORS AND AWARDS

- **Best Wiki Winner** and **Best Hardware & Target Design**, International Directed Evolution Competition (2023, 2024) *Top 1%*
- **Silver Medal**, International Genetically Engineered Machine Competition (iGEM) (2024) *Top 15%*
- **Meritorious Winner**, ICM Mathematical Modeling Competition (2023) *Top 6%*
- **Outstanding Student Scholarship**, Lanzhou University (2022, 2024) *Top 15%*